

CPT202 2023/2024 Semester 2
Final Report for Software Engineering Group Project

2023/2024 Semester 2

Fitness Trainer Appointment System

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<http://39.107.226.195:9418/H-XYZHomePage>

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Introduction

A fitness center's success heavily relies on an efficient booking system, as the traditional method of manual record-keeping often leads to errors, delays, and inefficiencies. Moreover, it can result in increased operating costs. Hence, we aim to develop a web-based appointment system for gyms that addresses these challenges and caters to evolving customer needs. This system will offer customers a convenient and user-friendly way to make bookings while enabling fitness centers to streamline their operations, reduce expenses, boost revenue generation, and enhance the overall experience for both members and staff.

The main goal of our project is to develop a user-friendly fitness center appointment system that automates the appointment process. User segmentation and specific objectives included:

Members: who have purchased a membership program and use the system to make reservations.

Member Functions: simplify the registration and login process; allow members to select classes, trainers and preferred time slots for booking; members can also view booking history and cancel.

Coaches: refers to a portion of the fitness center staff that is responsible for teaching classes.

Coach Functions: Coaches can easily access their accounts and view work schedules.

Managers: refers to another part of the fitness center staff who is responsible for administration.

Manager Functions: The manager has full visibility of all bookings in the system; manages the recharge system; efficiently manages trainer and class information; and provides viewing options for statistical business reports.

The system is based on four assumptions: A: Members are willing to provide personal information and agree to all system usage agreements. B: Members are willing to recharge their accounts through offline payments and use their account balances to purchase membership programs. C: Hired coaches are available during their working hours. D: Members are not able to change their membership plan during their active membership period, if they need to change, they need to wait until their current plan expires, this is verbally communicated to the member by the manager at the offline payment.

At the beginning of the project, I worked with the project owner to lead the project and develop the PBI design and naming convention. During the development period, I was responsible for managing and setting up the GitHub repository and providing version control, as well as the development of six main PBIs: These six PBIs, five of which are on the Manager side, include: View all appointment information, Recharge Member Accounts, Manager Dashboard weekly summary, Manager Dashboard Active member view, Manager Dashboard view Appointments Status. Also, logout on Manager, Member and Trainer sides. In the later stage of the project, I was mainly responsible for the writing of reports, the integration and revision of all documents, and the editing of demonstration videos.

In following the framework of Scrum development, this report will describe the development process of a fitness instructor booking system in terms of software development process, software design, change management, and legal, social, ethical and professional issues.

Software Development Process

The Scrum development process for our group is shown in Appendix A. Scrum makes projects easier to manage, allows teams to deliver high-quality work faster and more frequently, and gives teams

more flexibility to adapt to change[1]. We use it for planning upcoming Sprints by holding regular Sprint planning meetings. In addition, the developers involved in the relevant PBI hold daily station meetings and Sprint Review Meetings from time to time to the acceptance of Sprint deliverables and summarize and improve the development of subsequent Sprints. We use an intelligent document to manage the Sprint development process, which is developed by me and made available to all members (shown in Appendix A). I am responsible for six main PBIs: View all appointment information, Recharge Member Accounts, Manager Dashboard weekly summary, Manager Dashboard Active member view, Manager Dashboard Active member view and logout three sides. Except for the logout, the content I was responsible for was on the manager side, which needed to be completed after the membership and trainer development. Therefore, in Sprint 1, I only needed to complete the development of Logout and Manager View all Reservations, but since the team members were not very familiar with the development process in Sprint 1, the page design was slow, and after consulting with the Product Owner and holding a sprint review meeting, the Logout section that I was responsible for was moved to Sprint 2, and the same View all appointment information was also moved to Sprint 2 because the member side was not finished.

Moving on to Sprint 2. The first and highest priority task was to deal with the logout portion of the Sprint 1 delay, I developed the logout functionality for all sides to be able to logout at the end of using the system. After the end of logout I started developing the recharge member account feature which is an important feature related to the member purchase plan, I discussed with the developer who developed the purchase plan. Because the original plan was difficult to implement and the way of recharge needed to be changed, I discussed with the developer who checked the user information on the manager side and decided to put the user information view and recharge on one page. And it was decided that this PBI would continue to be developed through Sprint 3. At the same time, while closely following the progress of course booking on the member side, I also continued to develop View all appointment information on the manager side. After the Sprint review meeting and Sprint retrospective meeting, apart from recharge, my development work in Sprint 2 was accepted.

Going into Sprint 3, I was at the top of my to-do list, and all the basic functionality was done and ready for Dashboard development. First, I finished the recharge integration left over from Sprint 2. After fully communicating with the developer in charge of feature integration, it was decided that I would be in charge of developing the features and design of the entire Dashboard. I chose to develop three features in parallel: Manager Dashboard weekly summary, Manager Dashboard Active member view and Manager Dashboard view Appointments Status. After the Sprint Review Meeting and Sprint Retrospective Meeting, all my development in Sprint 3 was accepted.

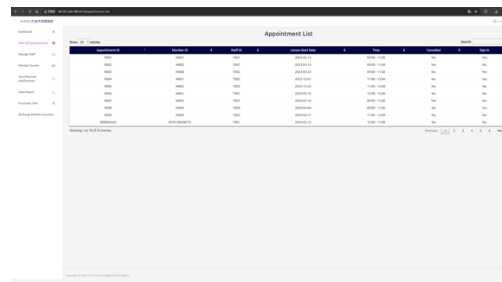
In three Sprints, I fully communicated with other developers and the Product Owner to enable agile development. Through Sprint planning meetings, daily station meetings, Sprint Review Meetings and Sprint Retrospective Meetings, our project was completed according to the established process.

Software Design

PBI: View all appointment information (All PBIs are described in Appendix D)

In this PBI, I first waited for the other developers responsible for developing the Member side of the course appointment to develop some of the content and then shared the *reservation* database with them. This page needs to show real-time appointments, as well as relevant information, and it also

needs to provide an easy-to-use page for the manager. The implementation page of this PBI is shown in Figure 1 and the sequence diagram involved is shown in Appendix B.



ID	Member Name	Course	Date	Time	Status
1	John Doe	Math	2023-10-01	10:00	Confirmed
2	Jane Smith	Science	2023-10-02	11:00	Confirmed
3	Mike Johnson	History	2023-10-03	09:00	Confirmed
4	Emily White	Art	2023-10-04	14:00	Confirmed
5	David Brown	Music	2023-10-05	15:00	Confirmed
6	Sarah Green	Physical Education	2023-10-06	08:00	Confirmed
7	Chris Black	Language	2023-10-07	13:00	Confirmed
8	Alexander Grey	Math	2023-10-08	10:00	Confirmed
9	Mia Blue	Science	2023-10-09	11:00	Confirmed
10	Noah Yellow	History	2023-10-10	09:00	Confirmed

Figure 1: Appointment List

The backend is implemented using the Spring Boot framework to handle business logic, accept front-end requests and interact with the database. **ReservationController.java** serves as the controller and contains a mapping method *getAllAppointments()* for GET requests. When the front-end accesses */api/appointments*, call *ReservationService*'s *findAll()* method to return a list of all appointment records. **ReservationService.java** acts as a service layer, responsible for business logic processing, and uses *ReservationRepository* to interact with the database. *findAll()* method retrieves all appointment records from the database. **ReservationRepository.java** as a repository layer, as an interface inherited from *JpaRepository*, providing basic CRUD operations. Through Spring Data JPA, automatically generate methods for common database operations. This session involved full communication with the developers who developed the booking sessions, shared and reviewed frequently and diligently through multiple stations and bi-daily online meetings to identify potentially undesirable discrepancies or issues.

The front-end part is relatively simple, without much optimization. The front-end is responsible for displaying the "View All Appointments" page and interacting with the user. It uses HTML templates, CSS for styling, and JavaScript for data acquisition and dynamic content generation. **view_all_schedule.html** implements this function, displays the necessary information, and enables the selection of how many appointments are displayed on a page, as well as the search and page flip functions. The search function uses JavaScript to listen to input events and re-filters and displays eligible appointment records each time the input changes. The display entry selection function uses JavaScript to listen to the change event, recalculate the paging and display the appointment records. JavaScript manages the paging button and listens to the click event to update the current page number when users click the paging button and redisplay the corresponding appointment record.

The database uses the *reservation* table, which has been previously created by other developers and will not be described here, we just need to query the contents. As a co-developer, I was also involved in reviewing the database design and raising issues for improvement. The database provides **Reservation.java** as an entity class that uses JPA (Java Persistence API) for object-relational mapping (ORM) to define the database table structure through the entity class. The Reservation entity class defines the database table structure for the appointment information, including the *id*, *memberName* and *course*. The Reservation entity class defines the database table structure of the reservation information, including *id*, *memberName*, *course*, *date*, and *time* fields, and uses the *@Entity* annotation to specify that this is a JPA entity class. the *ReservationRepository* performs a query operation to retrieve and return all reservation records to the service tier.

PBI: Dashboard (Manager's home page) (All PBIs are described in Appendix D)

For the sake of description, here I combine three PBIs: View all appointment information, Recharge Member Accounts, Manager Dashboard weekly summary, Manager Dashboard Active member view and Manager Dashboard Active member view into one Dashboard for overall description.

Manager Dashboard is used as the homepage of the manager's end, which needs to display some important management information in a concise and clear way, therefore, it needs more consideration on the front-end design. It needs to meet the characteristics of eye-catching and intuitive, business value, and provide help for gym management. The implementation of this page is shown in Figure 2, and the sequence diagram of the query data is shown in Appendix B.

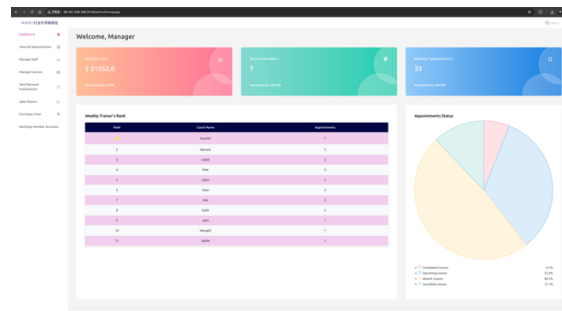


Figure 2: Dashboard (Manager homepage)

On the front end, this is done mainly through **manager_homepage.html**. Includes multiple sections: welcome message, weekly sales, active members, weekly appointments, coach rankings and appointment status charts. For aesthetics, I brought in several third-party libraries for implementation and used Bootstrap version 4.5.2 CSS files to provide predefined styles and layouts and ensure responsive design. The Material Design Icons icon library was used and the charting section used the *Chart.js* charting library to present the data. Also due to the established plan of displaying weekly data in the Dashboard and the need to compare it with the previous week's, I introduced *Moment.js* to provide date and time handling functionality, while the previously used *Chart.js* also supports the handling of charts with timelines. For the comparison between the current week and last week, after the acquisition of time, the comparison may appear to grow or decrease, by acquiring the symbol before the value to determine the growth or decrease, the value displayed is the absolute value acquired by the database. In this section, I followed the scrum framework and invited the developers responsible for the overall UI design of the system to participate in the review. Following their suggestions, I made some adaptations, mainly in terms of layout and color scheme. The back-end part is simpler, using the controller **StaffController.java** to handle the relevant HTTP requests. Get weekly sales, active members, weekly appointments, coach ranking data, and others. The database is operated using Spring Data JPA as before. There are multiple tables associated here, including member, purchase, reservation, etc., which were previously designed by other developers and now just need to execute queries. I followed the scrum and participated in the review. Since Dashboard needs to be associated with multiple databases, the **StaffServiceImpl.java** implements the **StaffService.java** interface for multiple queries. The database architecture is shown in Appendix C.

PBI: Recharge Member Accounts (All PBIs are described in Appendix D)

This feature is an integration of both Member Management and Recharge Member Accounts into one page. As it is difficult to implement an external payment interface, which requires a formal

application with an electronic payment provider (e.g., Alipay, WeChat), we do not have a business license and our website has not passed the ICP filing, so we set up that we can only pay the bill offline, and then the manager will add the balance to the user in the system. Such problems and the scrum handling by the development team is described in detail in the Change Management section. I briefly narrate through **adminManageMembers.html**, which displays the information of all the members, including member ID, name, phone number, plan and remaining amount, and provides the recharge function. Backend **MemberController.java** handles the recharge request, **MemberService.java** defines the interface for the recharge operation, **MemberServiceImpl.java** implements the lookup of the user and updates the balance, and **MemberRepository.java** interacts with the database to manage the member data. The database updates the *remaining_sum* column in the *member* table. The realized page is shown in Figure 3, and the corresponding sequence diagram is shown in Appendix B. When the vast majority of this feature was completed, I invited the developer viewing the user information to review it, reducing unnecessary workload.

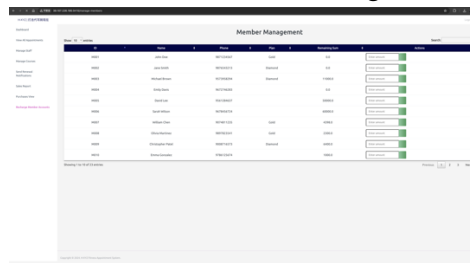


Figure 3: Recharge

PBI: Logout (All PBIs are described in Appendix D)

This PBI is important, but at the same time simple to implement. All three sides need the function of logout from the system, we define that clicking the button in the upper right corner can logout and then redirect to the login page. three html files, **manager-nav.html**, **member-nav.html** and **trainer-nav.html**, all carry out the logout implementation. Correspondingly, in the backend, **MemberController.java** and other three similar classes define the point where the logout request is handled, and after destroying the session, it is redirected to the login page. The implementation page is shown in Figure 4 and the activity diagram is shown in Appendix B. During this session, I had a number of station meetings with the developers of the login page and homepage to discuss and adapt the development to each other.

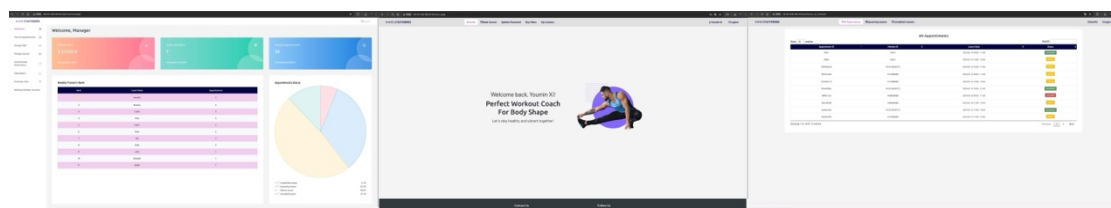


Figure 4: Logout

Change Management

In line with Scrum's core values of responding to change over following a plan, the requirements changed at some point in the development content I was responsible for. In Recharge Member Accounts, the initial idea was that we access a third-party payment platform to realize the payment, but after investigation, we found that the external payment interface is difficult to realize, and we

need to apply formally to the electronic payment providers (e.g., Alipay, WeChat), we don't have a business license, and the website didn't pass the ICP filing¹. Thus, we set up the payment to be made offline only and then the administrator will add the balance to the user in the system. After that, we decided to integrate this feature into the Member Management page. The integration will be added to the next Sprint. This decision was made mid-development after the issue was discovered, discussed directly with the Project Owner, and then brought up in our development team's bi-daily online progress meetings, where a unanimous decision was made to make the change.

Legal, social, ethical, and professional issues

In my development section, this part is not covered. I will consider it in the whole project.

Legal Issues: Mainly data protection and privacy: comply with privacy laws when collecting and storing users' personal info. Ensure service contract terms are clear and agreed upon during registration. Confirm legal responsibility for reservation issues or financial losses due to system failure when signing a development contract with the customer (gym). Engage professional legal counsel for support and regularly review and update the Privacy Policy and User Agreement. **Social Issues:** Proper handling and storage of users' health information to prevent misuse[2]. Use advanced encryption technology and data protection measures to secure user data. **Ethical Issues:** Ensure informed consent and respect for privacy when collecting and processing user information[3]. Transparency of system operations and data processing processes is required to ensure that users can understand how their data is being used. Avoid unfair treatment of some users due to algorithmic bias. In addition to advanced encryption technologies and data protection measures, these issues require clear and easy-to-understand user guides and privacy statements to help users understand and protect their rights. **Professional Issues:** Ensure the stability and reliability of the system and avoid frequent system crashes or failures. Provide timely and effective user support services to solve problems encountered by users in the process of use. Regularly update the system, fix vulnerabilities, optimize the user experience, and ensure that the system is always in the best operating condition. These issues require us to establish a system backup and recovery mechanism to prevent data loss and arrange for the development of the maintenance team after the launch.

Conclusion

In this development project, I gained a deep understanding of the Agile process and tested Scrum development in practice. After three rounds of Sprint, I learned front-end development and integrated it with back-end and database technologies. I also experienced the entire team development process, mastering how to explore business value in software projects and improving collaboration and communication skills. However, several improvements are necessary for our project to be viable. We need a more powerful server and a proper domain name, credentials for the payment platform, and mobile compatibility, as the current web version only supports computers. Additionally, the current features are basic; we need to develop specialized features for different gyms.

¹ <https://opendocs.alipay.com/open/204/105051?pathHash=b91b9616>

References

- [1] M. Kalenda, P. Hyna, and B. Rossi, "Scaling agile in large organizations: Practices, challenges, and success factors," *Journal of Software: Evolution and Process*, vol. 30, no. 10, p. e1954, 2018.
- [2] T. C. Rindfleisch, "Privacy, information technology, and health care," *Communications of the ACM*, vol. 40, no. 8, pp. 92-100, Aug. 1997.
- [3] Y. O'Connor, W. Rowan, L. Lynch, et al., "Privacy by design: informed consent and internet of things for smart health," *Procedia Computer Science*, vol. 113, pp. 653-658, 2017.

Appendix

Test Account (You can test our system with these accounts)

Member Phone: 13888888 Member Password: a12345

Manager Phone:74189470 Manager Password: Imanage1234

Trainer Phone:17792972 Trainer Password: IISdha17891

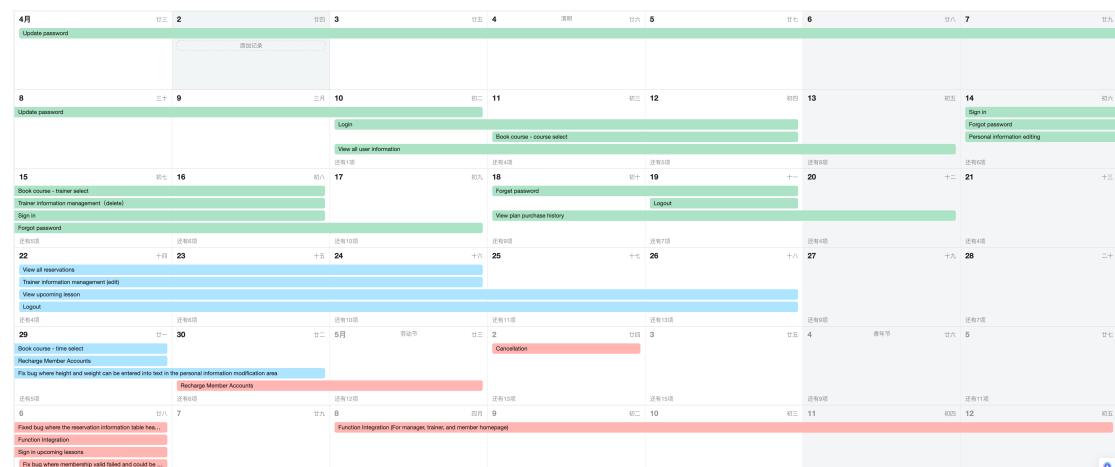
The detailed implementation code has been submitted in the team report, my part has not made any new modifications, please check the team report if necessary.

Appendix A: Sprint

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My character assignments for the three Sprint cycles

Yunzhou Zhong	Zhiqiang Zhang	Ruiyu Xu	Ying Zheng	Zhuo Wang	Tianqi Yang	Yueli Li	Ruihan Zhong	Jing Huang
View all reservation information User Role: Manager Deadline: 2024/4/21 Schedule: 42% PO Verification: PO Start Date: 2024/4/18	View all user information User Role: Manager Deadline: 2024/4/13 Schedule: 100% PO Verification: PO Start Date: 2024/4/10	Forgot password User Role: Member Deadline: 2024/4/17 Schedule: 100% PO Verification: PO Start Date: 2024/4/14	Book course - course select User Role: Member Deadline: 2024/4/12 Schedule: 100% PO Verification: PO Start Date: 2024/4/11	Login User Role: Member Deadline: 2024/4/12 Schedule: 100% PO Verification: PO Start Date: 2024/4/10	Update password User Role: Member Deadline: 2024/4/13 Schedule: 100% PO Verification: PO Start Date: 2024/4/11	Trainer information management (add) User Role: Manager Deadline: 2024/4/13 Schedule: 100% PO Verification: PO Start Date: 2024/4/11	Course information management (edit) User Role: Manager Deadline: 2024/4/19 Schedule: 100% PO Verification: PO Start Date: 2024/4/13	Personal information viewing User Role: Member Deadline: 2024/4/18 Schedule: 100% PO Verification: PO Start Date: 2024/4/13
View all reservation information User Role: Manager Deadline: 2024/4/20 Schedule: 100% PO Verification: PO Start Date: 2024/4/22	View plan purchase history User Role: Manager Deadline: 2024/4/20 Schedule: 100% PO Verification: PO Start Date: 2024/4/18	View upcoming lesson User Role: Member Deadline: 2024/4/28 Schedule: 100% PO Verification: PO Start Date: 2024/4/22	Reservation function navigation bar User Role: Member Deadline: 2024/5/3 Schedule: 100% PO Verification: PO Start Date: 2024/5/1	Sign in User Role: Member Deadline: 2024/4/16 Schedule: 100% PO Verification: PO Start Date: 2024/4/14	Send marketing messages to members whose expiration date is within 15 days User Role: Manager Deadline: 2024/4/21 Schedule: 100% PO Verification: PO Start Date: 2024/4/17	Trainer information management (search) User Role: Manager Deadline: 2024/4/26 Schedule: 100% PO Verification: PO Start Date: 2024/4/24	Course information management (add) User Role: Manager Deadline: 2024/5/3 Schedule: 100% PO Verification: PO Start Date: 2024/4/23	Personal information editing User Role: Member Deadline: 2024/4/19 Schedule: 42% PO Verification: PO Start Date: 2024/4/14
Recharge Member Accounts User Role: Manager Deadline: 2024/4/29 Schedule: 100% PO Verification: PO Start Date: 2024/4/26	Manager's Sales Report User Role: Manager Deadline: 2024/4/28 Schedule: 100% PO Verification: PO Start Date: 2024/4/17	Forgot password User Role: Trainer Deadline: 2024/4/19 Schedule: 100% PO Verification: PO Start Date: 2024/4/18	Fix bug where membership valid failed and could be accessed from other pages through the top navigation bar User Role: Member Deadline: 2024/5/8 Schedule: 100% PO Verification: PO Start Date: 2024/5/6	Login User Role: Trainer Deadline: 2024/4/14 Schedule: 100% PO Verification: PO Start Date: 2024/4/12	View upcoming lessons User Role: Trainer Deadline: 2024/4/26 Schedule: 100% PO Verification: PO Start Date: 2024/4/24	Function integration User Role: Trainer Deadline: 2024/5/8 Schedule: 100% PO Verification: PO Start Date: 2024/4/29	Add course attribute - is_active (instead of deleting course) User Role: Manager Deadline: 2024/5/10 Schedule: 100% PO Verification: PO Start Date: 2024/5/5	Fix bug where height and weight can be entered into text in the personal information modification area User Role: Member Deadline: 2024/4/20 Schedule: 100% PO Verification: PO Start Date: 2024/4/25
Logout User Role: Member Trainer Manager Deadline: 2024/4/26 Schedule: 100% PO Verification: PO Start Date: 2024/4/26	Trainer view personal information User Role: Trainer Deadline: 2024/5/8 Schedule: 100% PO Verification: PO Start Date: 2024/5/6	Update the forgot password function User Role: Member Deadline: 2024/5/8 Schedule: 100% PO Verification: PO Start Date: 2024/5/6	The front-end page formatting unfile User Role: Member Deadline: 2024/5/8 Schedule: 100% PO Verification: PO Start Date: 2024/5/6	Page design User Role: Trainer Deadline: 2024/4/21 Schedule: 100% PO Verification: PO Start Date: 2024/4/21	View completed lessons User Role: Trainer Deadline: 2024/4/28 Schedule: 100% PO Verification: PO Start Date: 2024/4/28	Sign in upcoming lessons User Role: Trainer Deadline: 2024/4/28 Schedule: 100% PO Verification: PO Start Date: 2024/4/28	Personal information editing User Role: Member Deadline: 2024/4/28 Schedule: 100% PO Verification: PO Start Date: 2024/4/28	View all appointment User Role: Member Deadline: 2024/4/28 Schedule: 100% PO Verification: PO Start Date: 2024/4/28

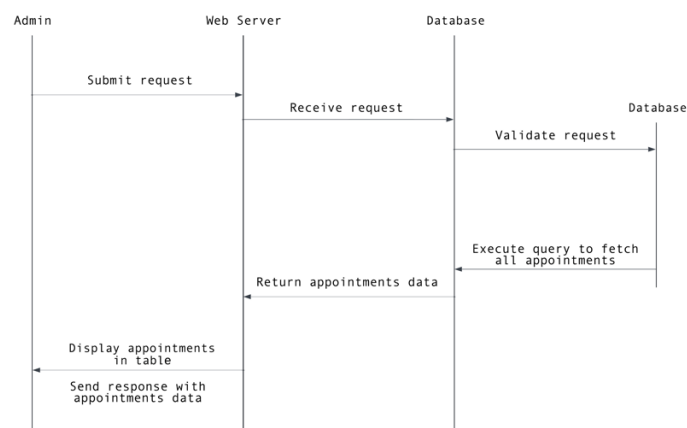


Our Scrum Management Intelligence Documentation

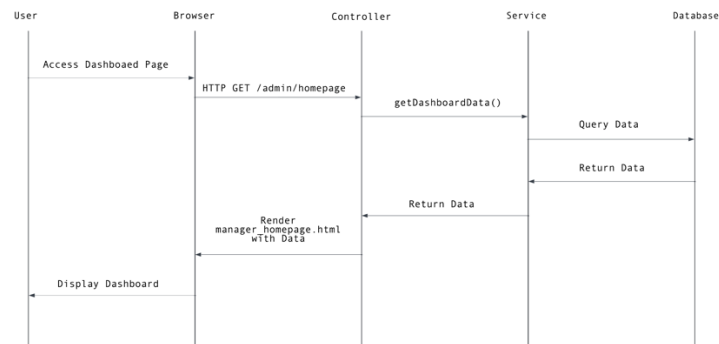
	Sprint	File Details	User Role	User Name	Week	Start Date	Deadline	State	Progress	Remarks	PO Verifies
3	Sprint 1	Forgot password	Member	Ruiyu Xu	7	2024/4/10	2024/4/15	Not Completed	40%	Requirement change	Done
4	Sprint 1	Personal information viewing	Member	Jing Huang	7	2024/4/8	2024/4/14	Completed	100%		Done
5	Sprint 1	Personal information editing	Member	Jing Huang	8	2024/4/14	2024/4/19	Not Completed	40%	Continue to complete in Sprint 2	Done
6	Sprint 1	Update password	Member	Tianci Yang	7	2024/4/10	2024/4/17	Completed	100%		Done
7	Sprint 1	Book course - course select	Member	Ying Zheng	7	2024/4/11	2024/4/14	Completed	100%		Done
8	Sprint 1	Book course - trainer select	Member	Ying Zheng	8	2024/4/15	2024/4/18	Completed	100%		Done
9	Sprint 1	Book course - time select	Member	Ying Zheng	8	2024/4/19		Not Completed	30%	Continue to complete in Sprint 2	Done
10	Sprint 1	Book course - confirmation	Member	Ying Zheng				Not Completed	0%	Postponed to Sprint 2	Done
11	Sprint 1	Login	Trainer	Ziwei Wang	7	2024/4/12	2024/4/14	Completed	100%		Done
12	Sprint 1	Course information management (view, edit)	Manager	Huohan Zhong	7	2024/4/8	2024/4/17	Completed	100%		Done
13	Sprint 1	Forgot password	Trainer	Ruiyu Xu	7	2024/4/10	2024/4/19	Completed	100%		Done
14	Sprint 1	View all user information	Manager	Zhiqiang Zhang	7	2024/4/10	2024/4/13	Completed	100%		Done
15	Sprint 1	View all reservation information	Manager	Yunzhou Zhong	7	2024/4/8	2024/4/21	Not Completed	40%	Continue to complete in Sprint 2	Done
16	Sprint 1	View plan purchase history	Manager	Zhiqiang Zhang	7	2024/4/18	2024/4/20	Completed	100%	Finish	Done
17	Sprint 1	Trainer information management (view)	Manager	Yiwei Li	7	2024/4/11	2024/4/13	Completed	100%		Done
18	Sprint 1	Trainer information management (add)	Manager	Yiwei Li	8	2024/4/17	2024/4/20	Completed	100%		Done
19	Sprint 1	Trainer information management (delete)	Manager	Yiwei Li	8	2024/4/17	2024/4/20	Not Completed	0%	delete this pbi, add defective to adjust trainer activation status instead of deleting	Done
20	Sprint 1	Page design	Navigation	Ziwei Wang	8	2024/4/16	2024/4/21	Completed	100%		Done
21	Sprint 1	Registration agreement	Navigation	Ziwei Wang	8	2024/4/17	2024/4/18	Completed	100%		Done
22	Sprint 1	Loginout	Member Trainer Manager	Yunzhou Zhong				Not Completed	0%	Postponed to Sprint 2	Done
23	Sprint 1	Member buy plan	Member	Zhiqiang Zhang	7	2024/4/10	2024/4/19	Not Completed	80%	Add limit about the plan type and add information in agreement in sprint2	Done
24	Sprint 1	Member buy plan (date update)	Member	Zhiqiang Zhang	8						
25	Sprint 1	Member buy plan(type update)	Member	Zhiqiang Zhang	8						
26	Sprint 2	Book course - time select	Member	Ying Zheng	9	2024/4/23	2024/4/29	Completed	100%	Fix bug where members could book classes for the day that has passed	Done
27	Sprint 2	Book course - confirmation	Member	Ying Zheng	10	2024/4/30	2024/5/3	Completed	100%		Done
28	Sprint 2	View upcoming lesson	Member	Ruiyu Xu	8	2024/4/22	2024/4/28	Completed	100%		Done
29	Sprint 2	View completed lesson	Member	Ruiyu Xu	8	2024/5/1	2024/5/3	Completed	100%		Done
30	Sprint 2	View all appointment	Member	Jing Huang	9	2024/4/22	2024/4/27	Not Completed	80%	Continue to complete in Sprint 3	Done
31	Sprint 2	Fix bug where height and weight can be entered into text in the personal information modification...	Member	Jing Huang	10	2024/4/25	2024/4/30	Completed	100%		Done
32	Sprint 2	Personal information editing (Front beautification)	Member	Jing Huang	10	2024/4/22	2024/5/5	Completed	100%		Done
33	Sprint 2	Course information management (add)	Manager	Huohan Zhong	9	2024/4/23	2024/4/28	Completed	100%		Done
34	Sprint 2	Personal information editing	Trainer	Huohan Zhong	10	2024/4/29	2024/5/3	Completed	100%		Done
35	Sprint 2	Trainer information management (add)	Manager	Yiwei Li	9	2024/4/22	2024/4/24	Completed	100%		Done
36	Sprint 2	Trainer information management (search)	Manager	Yiwei Li	10	2024/4/29	2024/5/2	Completed	100%		Done
37	Sprint 2	Trainer information management (Front beautification)	Manager	Yiwei Li	10	2024/4/29	2024/5/2	Completed	100%	also realize the pagination and ascending and descending order functions	Done
38	Sprint 2	View expired member information	Manager	Tianci Yang	9	2024/4/26	2024/4/28	Not Completed	80%	Page text display error	Done
39	Sprint 2	Send marketing messages to members whose expiration date is within 15 days	Manager	Tianci Yang	10	2024/4/29	2024/5/3	Completed	100%		Done
40	Sprint 2	Manager's Sales Report	Manager	Zhiqiang Zhang	9	2024/4/17	2024/4/28	Completed	100%	Add explore	Done
41	Sprint 2	Manager's Sales Report(Display)	Manager	Zhiqiang Zhang	9	2024/4/17	2024/4/28	Completed	100%		Done
42	Sprint 2	View all reservation information	Manager	Yunzhou Zhong	9	2024/4/22	2024/5/3	Completed	100%		Done
43	Sprint 2	Recharge Member Accounts	Manager	Yunzhou Zhong	9	2024/4/25	2024/4/28	Completed	100%	Needs to be reintegrated into the Member Management page	Done
44	Sprint 2	View upcoming lessons	Trainer	Tianci Yang	9	2024/4/22	2024/4/24	Completed	100%		Done
45	Sprint 2	View completed lessons	Trainer	Tianci Yang	9	2024/4/22	2024/4/24	Completed	100%		Done
46	Sprint 2	View all reservations	Trainer	Tianci Yang	9	2024/4/22	2024/4/24	Completed	100%		Done
47	Sprint 2	The front-end page formatting unfile	Member Trainer Manager	Ying Zheng Ziwei Wang	10	2024/5/3		Not Completed	70%	Continue to complete in Sprint 3	Done
48	Sprint 2	Reservation function navigation bar	Member	Ying Zheng	10	2024/5/1	2024/5/3	Completed	100%		Done
49	Sprint 2	Loginout	Member Trainer Manager	Yunzhou Zhong	9	2024/4/22	2024/4/28	Completed	100%		Done
50	Sprint 3	Fix bug where members could book classes for the day that has passed	Member	Ying Zheng	11	2024/5/5	2024/5/6	Completed	100%		Done
51	Sprint 3	Fix bug where membership valid failed and could be accessed from other pages through the top...	Member	Ying Zheng	11	2024/5/6	2024/5/6	Completed	100%		Done
52	Sprint 3	Update the forgot password function	Member	Ruiyu Xu	9	2024/5/5	2024/5/11	Completed	100%		Done
53	Sprint 3	Cancellation	Member	Ruiyu Xu	11	2024/5/13	2024/5/15	Completed	100%		Done
54	Sprint 3	View all appointment	Member	Jing Huang	11	2024/5/6	2024/5/9	Completed	100%		Done
55	Sprint 3	Fixed bug where the reservation information table header was incorrect	Trainer	Tianci Yang	11	2024/5/6	2024/5/7	Completed	100%		Done
56	Sprint 3	The front-end page formatting unfile	Member Trainer Manager	Ying Zheng Ziwei Wang	11	2024/5/3	2024/5/8	Completed	100%		Done
57	Sprint 3	Function integration	Trainer	Tianci Yang Yiwei Li	12	2024/5/13	2024/5/15	Completed	100%		Done
58	Sprint 3	Sign in upcoming lessons	Member	Yiwei Li	11	2024/5/6	2024/5/9	Not Completed	0%	delete this pbi and add the pbi of trainer sign in, transfer the sign in task from member to trainer	Done
59	Sprint 3	Sign in upcoming lessons	Trainer	Yiwei Li	11	2024/5/6	2024/5/9	Completed	100%		Done
60	Sprint 3	Function integration (for manager, trainer, and member homepage)	Manager Member Trainer	Ying Zheng	11	2024/5/8	2024/5/11	Completed	100%		Done
61	Sprint 3	Manager Dashboard weekly summary	Manager	Yunzhou Zhong	10	2024/5/2	2024/5/15	Completed	100%		Done
62	Sprint 3	Manager Dashboard Active member view	Manager	Yunzhou Zhong	10	2024/5/2	2024/5/15	Completed	100%		Done
63	Sprint 3	Manager Dashboard Active member view	Manager	Yunzhou Zhong	10	2024/5/2	2024/5/15	Completed	100%		Done
64	Sprint 3	Recharge Member Accounts	Manager	Yunzhou Zhong	10	2024/4/30	2024/5/11	Completed	100%		Done
65	Sprint 3	Add course attribute - is_active (instead of deleting course)	Manager	Huohan Zhong	11	2024/5/6	2024/5/9	Completed	100%		Done
66	Sprint 3	Trainer view personal information	Trainer	Zhiqiang Zhang	10	2024/5/5	2024/5/8	Completed	100%		Done
67	Sprint 3	Manager's Sales Report(Explore)	Manager	Zhiqiang Zhang	11	2024/5/3		Completed	100%		Done
68	Sprint 3	Popup fade time	Member Trainer Manager	Ziwei Wang	10	2024/5/3	2024/5/5	Completed	100%		Done

Scrum Development Process

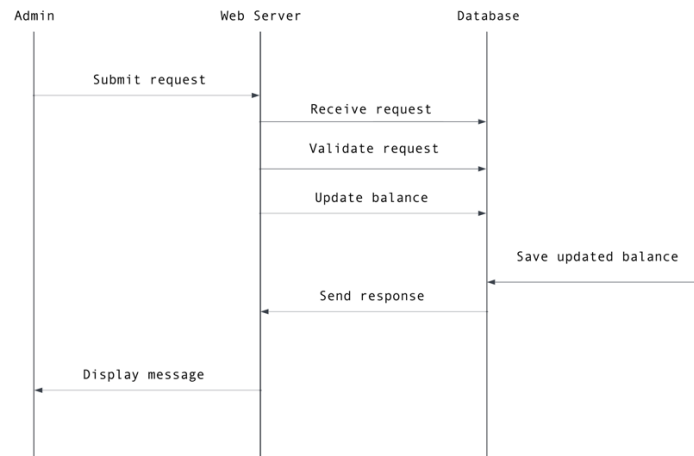
Appendix B: UML



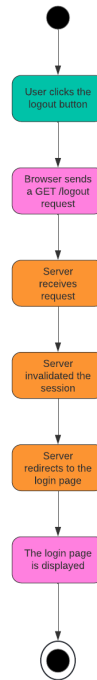
Sequence Diagram of view all appointment information PBI



Sequence Diagram of Dashboard PBI

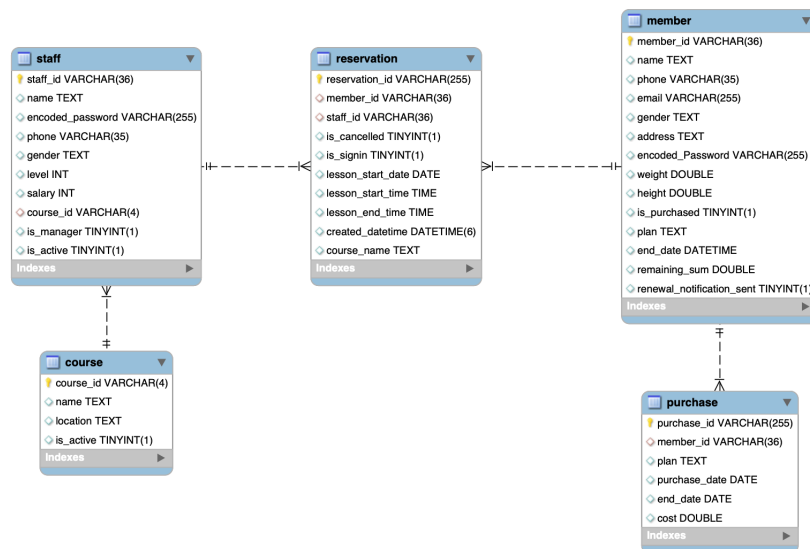


Sequence Diagram of Recharge PBI



Activity Diagram of Logout PBI

Appendix C: Database ER Diagram



Appendix D: PBIs

Product Backlog Item

PBI #	1
Title	View All Appointment Information

Creation Date	April 1, 2024	Status	New	Sprint	1+2
Priority	Important	Effort		Business Value	

Story

As a manager, I want to view all appointment information so that I can effectively manage and oversee the appointments made by members, ensuring that the schedules are correctly maintained and any potential conflicts or issues are identified promptly.

Description

1. The system should provide a comprehensive list of all appointments, including details such as member name, course, date, and time.
2. The page must be user-friendly and allow managers to easily navigate through the list of appointments.
3. The list should be displayed in real-time, reflecting any updates or changes made.
4. The system should allow filtering and searching of appointments based on various criteria such as member name, course, date, and time.
5. It should include pagination to manage the display of a large number of appointments.
6. The appointment list should be sortable by different columns (e.g., by date, by member name).

Acceptance Criteria

- 1) Given that I am a valid manager, when I navigate to the “View All Appointment Information” page, then I should see a list of all appointments with details such as member name, course, date, and time.
- 2) Given that I am on the “View All Appointment Information” page, when I enter a search query and click the search button, then the system should display appointments matching the search criteria.
- 3) Given that I am on the “View All Appointment Information” page, when I select a column header to sort by, then the list should be reordered accordingly.
- 4) Given that I am on the “View All Appointment Information” page, when I click on an appointment entry, then I should be directed to the detailed view page for that appointment.
- 5) Given that I am on the “View All Appointment Information” page, when the number of appointments exceeds the display limit, then the system should provide pagination controls to navigate through the list.

Grooming

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Discussion

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Related PBIs

Book course - course select
Book course - trainer select
Book course - time select
Book course - confirmation

Product Backlog Item

PBI #	2
Title	Recharge Member Accounts

Creation Date	April 1, 2024	Status	New	Sprint	2+3
Priority	Important	Effort		Business Value	

Story

As a manager, I want to recharge member accounts so that members can use their account balance to book appointments and services.

Description

1. The system should provide an interface for managers to add balance to member accounts. 2. The manager should be able to search for a member by name or ID. 3. The system should validate the recharge amount and update the member's balance accordingly.
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Acceptance Criteria

1) Given that I am a valid manager, when I navigate to the "Recharge Member Accounts" page, then I should see a form to enter member ID and recharge amount. 2) Given that I am on the "Recharge Member Accounts" page, when I enter a valid member ID and recharge amount and submit the form, then the system should update the member's balance and display a confirmation message. 3) Given that I am on the "Recharge Member Accounts" page, when I enter an invalid member ID or recharge amount and submit the form, then the system should display an error message indicating the issue. 4) Given that I successfully recharge a member's account, when I navigate to the member's profile, then I should see the updated balance.

Grooming

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Discussion

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Related PBIs

Member Management Page

Product Backlog Item

PBI #	3
Title	Dashboard Weekly Summary

Creation Date	April 1, 2024	Status	New	Sprint	3
Priority	Important	Effort		Business Value	

Story

As a manager, I want to see a weekly summary of key metrics on the dashboard so that I can monitor the performance and activity of the fitness center.
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Description

1. The dashboard should display a summary of weekly metrics such as total appointments, new member registrations, and revenue. 2. The summary should be presented in an easy-to-read format with charts and graphs where applicable. 3. The data should be updated in real-time to reflect the latest information.
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Acceptance Criteria

1) Given that I am a valid manager, when I navigate to the dashboard, then I should see a summary of weekly metrics including total appointments, new member registrations, and revenue. 2) Given that I am on the dashboard, when the data is updated, then the summary should reflect the latest information in real-time. 3) Given that I am on the dashboard, when I hover over or click on a metric, then I should see additional details or a drill-down view of the data.
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Grooming

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Discussion

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Related PBIs

Manager Dashboard Active Member View
Manager Dashboard view Appointments Status

Product Backlog Item

PBI #	4
Title	Manager Dashboard Active Member View

Creation Date	April 1, 2024	Status	New	Sprint	3
Priority	Important	Effort		Business Value	

Story

As a manager, I want to view the active members on the dashboard so that I can monitor member engagement and activity.

Description

1. The dashboard should display a summary of active members, including their recent activity and current status.
2. The data should be updated in real-time to reflect the latest member activity.

Acceptance Criteria

- 1) Given that I am a valid manager, when I navigate to the dashboard, then I should see a summary of active members with their recent activity and current status.
- 2) Given that I am on the dashboard, when the data is updated, then the summary should reflect the latest member activity in real-time.

Grooming

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Discussion

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Related PBIs

Dashboard Weekly Summary
Manager Dashboard view Appointments Status

Product Backlog Item

PBI #	5
Title	Manager Dashboard view Appointments Status

Creation Date	April 1, 2024	Status	New	Sprint	3
Priority	Important	Effort		Business Value	

Story

As a manager, I want to view the status of appointments on the dashboard so that I can monitor the booking trends and identify any issues that need attention.

Description

1. The dashboard should display the status of appointments, including the number of booked, cancelled, and completed appointments.
2. The information should be presented in a graphical format, such as charts or graphs.
3. The data should be updated in real-time to reflect the latest status of appointments.

Acceptance Criteria

- 1) Given that I am a valid manager, when I navigate to the dashboard, then I should see a graphical representation of the status of appointments, including booked, cancelled, and completed appointments.
- 2) Given that I am on the dashboard, when the data is updated, then the graphical representation should reflect the latest status of appointments in real-time.

Grooming

Discussion

Related PBIs

Dashboard Weekly Summary
Manager Dashboard Active Member View

Product Backlog Item

PBI #	6
Title	Logout

Creation Date	April 1, 2024	Status	New	Sprint	1+2
Priority	Important	Effort		Business Value	

Story

As a user, I want to securely log out of the system so that my account remains protected when I am not using it.

Description

1. The system should provide a logout button that is easily accessible from any page.
2. When the logout button is clicked, the user should be logged out and redirected to the login page.

Acceptance Criteria

- 1) Given that I am a logged-in user, when I click the logout button, then I should be logged out and redirected to the login page.
- 2) Given that I am logged out, when I attempt to navigate to a protected page, then I should be redirected to the login page.
- 3) Given that I am logged out, when I log in again, then my previous session should not be restored, and I should start a new session.

Grooming

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Discussion

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Related PBIs

Login Homepage
